

The Art of Queueing

Reinforcement Learning for visitor flow at the Gallerie degli Uffizi in Florence

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Abstract

The Uffizi Gallery makes congestion visible as a reinforcement-learning problem: capacity is not scarce everywhere, but demand concentrates around Botticelli, Leonardo, Raphael-Michelangelo and Caravaggio. We use two linked model layers. A minute-step agent-based simulator moves heterogeneous visitors through the room graph and evaluates institutional interventions under a revenue-preservation constraint. On top of those simulated crowds, RL agents solve the visitor problem: when to stay, when to move and how far ahead, under RAMA, to reserve masterpiece slots. The technical work is mostly formulation. Appreciation progress makes dwell decisions Markovian; dense closing pressure prevents late trapping; reward design discounts congestion without making masterpieces unattractive; action masks keep PPO on legal graph and booking-phase choices. In the final simulator evaluation, the all-11 bundle raises completed welfare by 31% at 5,000 visitors with revenue essentially preserved. Five-seed MaskablePPO booking policies beat matched no-booking baselines across profiles and crowds in every cell.

1 Problem and Modeling Strategy

The project starts from a common-pool resource problem. The Uffizi has enough total floor area to serve many visitors, but visitor utility is degraded when a large share of demand arrives in the same rooms at the same time. The clearest bottleneck is the Botticelli chain: rooms A11 and A12 lie on a forced path and contain *Primavera* and *The Birth of Venus*. Leonardo and Raphael/Michelangelo generate secondary peaks, while many other galleries remain underused. The RL question is therefore not simply whether an agent can find a shortest path. It is whether a learner can use time, route choice and reservation lead time to obtain a better visit than a fixed itinerary.

The implementation separates two decision problems. At the museum level, a minute-step simulator runs a full day of heterogeneous visitors under the legal 900-person capacity constraint. Interventions alter arrival curves, access rules, prices and secondary attractions; they are evaluated by whether they improve welfare without making revenue politically infeasible. At the individual level, one controlled visitor optimizes a visit conditional on either the status-quo or intervened crowd process. In the status quo, the relevant action is mostly whether to remain in a room or continue along a planned route. Under RAMA, the decisive action happens before entry: the visitor chooses how far in advance to reserve masterpiece slots and must then pace the walk to reach those windows.

This distinction is important because the simulator and the RL agents answer different questions. The simulator is the population transition model and produces the museum-wide welfare and revenue statistics. The RL agents estimate an optimized individual ceiling under a fixed population process. Collapsing the two would obscure the difference between evaluating an institutional intervention and solving a single visitor’s sequential decision problem.

2 The Virtual Uffizi

2.1 Graph and calibration

The physical museum is encoded as a directed graph $G = (V, E)$ built from the 98 rooms emphasized in the slides plus operational nodes such as the entry, staircases, terrace and exit. Each room carries four central attributes: cultural importance, dwell-time magnetism, legal or effective capacity and section. Ordinary doors are represented bidirectionally; staircases and exit routes are directed. The topology is validated with structural invariants: weak connectivity, reachability from the entry, reachability of an exit from every gallery, high degree for the main corridor hubs and a forced Botticelli chain A10–A11–A12–A13.

Visitors arrive through 37 fifteen-minute entry slots with a broad morning-weighted envelope. The simulator uses a one-minute time step, a 615-minute baseline day (08:15–18:30) and a final-entry cutoff at 17:30. The full intervention scenario extends the day to 720 minutes. The population mix is 60% Instagram tourists, 30% normal tourists and 10% art lovers. Instagram tourists concentrate on famous rooms and the panoramic terrace, move quickly and are nearly crowd-indifferent. Normal tourists visit the famous rooms plus a few canonical additions, such as the Tribune and Caravaggio. Art lovers have heterogeneous taste vectors, longer dwell times, stronger crowd aversion and a higher probability of accepting alternative routes.

The movement model is local and stochastic. Conditional on deciding to move, a visitor chooses among neighbors with weights of the form

$$w_{i \rightarrow j} = \max\{0.01, b_{\text{route}}(j) + b_{\text{importance}}(j) + b_{\text{crowd}}(j)\}, \quad P(i \rightarrow j) = \frac{w_{i \rightarrow j}}{\sum_{k \in N(i)} w_{i \rightarrow k}}.$$

The route term rewards the next room on the recommended itinerary, the importance term pulls visitors toward personally valuable rooms and the crowd term favors low-density alternatives only for crowd-sensitive visitors or information-guided interventions. This expression is only the schematic core. The executable simulator also contains policy-specific movement logic for RAMA slots, low-budget exit seeking, tour groups, Caravaggio priority and masterpiece bypasses. The model is intentionally behavioral rather than globally optimal: the simulator is designed to reproduce plausible crowd pressure, while the controlled RL environments solve the individual optimization problem separately.

2.2 Museum interventions

The intervention catalog contains access controls, price instruments, time architecture and demand redistribution. The slide language calls the final scenario “all 11” because the publication catalog has eleven intervention specifications, including two pre-bundled combinations. In code, these specifications are merged with last-write-wins semantics into nine distinct active switches: RAMA, secondary-attractor enrichment, extended hours, dynamic pricing, resident annual pass expansion, timed entry, quiet hours, a tour-group cap of 10 and a EUR150 group surcharge. A separate portfolio optimizer screens interventions individually and then applies greedy forward selection followed by one-step local search on a scalarized welfare/revenue objective. That optimization is useful as an economic robustness check, but the RL booking agents are trained in the all-11 world because it is the environment in which RAMA access is active and internally consistent.

Figure 1 gives the corresponding spatial evidence. At the 5,000-visitor crowd, Botticelli peak density falls from 1.82 to 1.00 and traffic is distributed over a longer operating window. The intervention does not make the museum empty; rather, it keeps the highest-demand rooms closer to usable capacity.

Table 1: Museum-wide simulator results: status quo versus the all-11 intervention world. Welfare is total completed visitor welfare. Revenue is daily ticket revenue in EUR.

Crowd	Welfare base	Welfare int.	Gain	Revenue base	Revenue int.
500	277,256	343,507	+24%	11,542	11,317
2,500	1,106,175	1,274,045	+15%	56,604	56,544
5,000	1,514,458	1,986,425	+31%	96,502	94,792

Table 2: The three MDP families used in the project.

Setting	State	Action	Algorithms
Toy museum	Discrete 5-tuple: room, time bin, Botticelli crowd bin, Leonardo crowd bin, visited mask	Stay or move to a neighbor	Q-learning, Double-Q
Status-quo walk	8 real features: drain, importance, density, on-target, plan share, time, egress, distance	Stay or move on	PPO, DQN
RAMA booking	25 real features: booking phase, lead screen, held windows, check-ins, access read, arrival priors, room tail	Phase-dependent lead, commit, stay, move choices	MaskablePPO, PPO, DQN

3 MDP Formulations

All individual agents are standard discounted Markov decision processes $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$, but the relevant state and action space change with the decision problem. The implementation lesson is that algorithm choice was secondary to specifying what the agent could observe and what decision it was actually allowed to make.

3.1 Tabular proof of concept

The toy museum has 12 rooms and deterministic crowd waves. It is small enough for a literal action-value table. The state is

$$s = (\text{room}, t_{\text{bin}}, c_{\text{Bott}}, c_{\text{Leo}}, m_{\text{visited}}),$$

where c_{Bott} and c_{Leo} are four-level crowd bins and m_{visited} is a bitmask over must-see rooms. Q-learning applies the masked Bellman update

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a' \in \mathcal{A}(s')} Q(s', a') - Q(s, a) \right],$$

where $\mathcal{A}(s')$ contains only legal graph moves. The toy confirms the basic economic idea: learning beats fixed rules because it discovers temporal arbitrage, visiting Botticelli off-peak rather than mechanically following the route. Over the five reported training seeds, Q-learning reaches 83.3 ± 3.8 mean return and Double-Q reaches 83.1 ± 3.7 , while the handcrafted rules range from -72.0 to 35.4 . The small Q versus Double-Q gap is not surprising: the toy is mostly deterministic, so there is little noisy maximization bias for Double-Q to remove.

3.2 Status-quo navigation

The status-quo full-museum agent uses a planned-route wrapper. It does not choose arbitrary neighboring rooms; instead, it decides whether to remain in the current target or continue to the

Crowd density heatmap: status quo vs all 11 interventions (~5000 visitors). Shared colour scale.

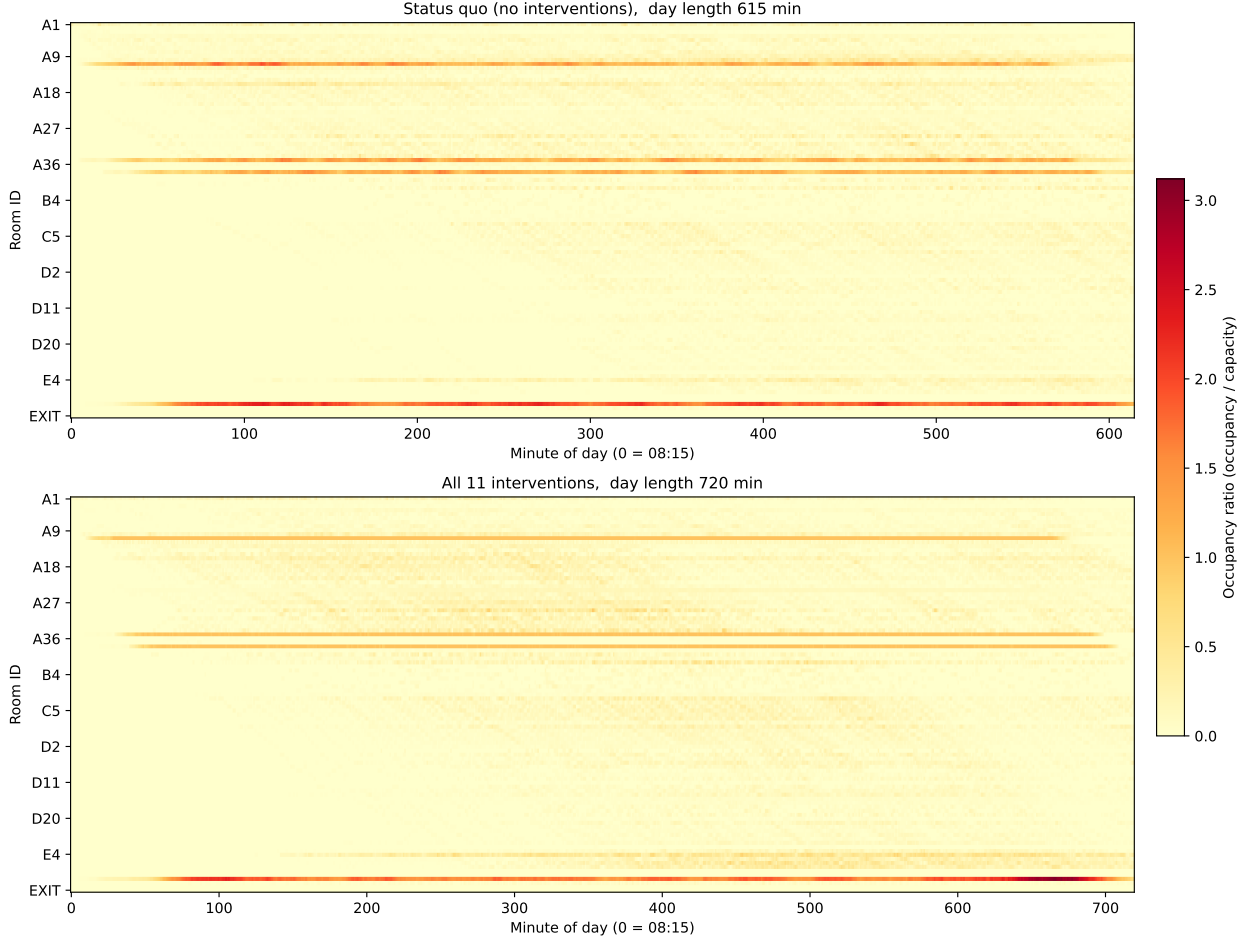


Figure 1: Crowd density under the status quo and under the full intervention bundle. The color scale is shared, so cooling of the red bands corresponds to a real reduction in occupancy over capacity.

next room in the planned walk. This gives the booking agent a fair comparison: the baseline is optimized, but it cannot use RAMA. The observation vector is

$$o_{\text{walk}} \in \mathbb{R}^8 = (\text{drain}, \text{importance}, \rho, \text{on-target}, \text{plan-share}, t, \text{egress}, \text{distance-to-next}).$$

The “drain” feature is crucial. Without an appreciation-progress state, the decision “stay until I have seen enough, then leave” is partially observed; the same physical room and time could require opposite actions depending on how long the visitor has already been absorbing the work.

3.3 RAMA as a two-phase MDP

The RAMA environment is the centerpiece. It wraps the same core museum reward, but changes the action interface. Phase 1 is a booking screen: choose a lead time from

$$\ell \in \{1, 7, 21, 35\} \text{ days},$$

then commit to masterpiece windows. Phase 2 is the walk, where the visitor stays or moves and must arrive inside the booked windows. The 25-dimensional observation groups into (i) booking position, (ii) slots open by lead, (iii) windows and check-ins held, (iv) access information for the current room, (v) frozen expected arrival times and (vi) the local room tail from the walk baseline.

The simplified availability rule is crowd-dependent:

$$\text{available}(g, \ell) = \mathbb{1} \left[\ell \geq L_{\max} p_g \frac{N_{\text{day}}}{2500} \right],$$

where p_g is the popularity of masterpiece group g and N_{day} is daily demand. This is the mechanism that makes late booking fail on busy days. Importantly, the policy is not told “book early when busy.” It only sees slot availability and downstream reward.

4 Reward Engineering

The deep agents share a core per-minute reward implemented in the full Uffizi environment:

$$r_t = r_{\text{art}} + r_{\text{time}} + r_{\text{close}} + r_{\text{complete}} + r_{\text{bored}} + r_{\text{checkin}} + r_{\text{wait}}.$$

The main term is art value with crowd discount and satiation:

$$r_{\text{art}} = I_i \frac{1}{1 + \alpha \rho_i^2} \text{clip} \left(\frac{D_i - e_i}{s}, 0, 1 \right), \quad D_i = \max\{1, k_d(I_i - I_0)\}.$$

Here I_i is the visitor-specific room importance, ρ_i is density over capacity, e_i is appreciation already extracted and D_i is ideal dwell time. Appreciation accumulates more slowly in crowded rooms, so congestion hurts twice: it lowers instantaneous utility and consumes more of the fixed day.

The final reward is the result of several failed intermediate specifications. A purely terminal penalty for not exiting produced an almost flat reward landscape and the agent often learned to remain in one room. The fix was to replace that sparse penalty with a dense closing signal,

$$r_{\text{close}} = -\kappa \max(0, d_{\text{exit}} + b - \tau),$$

where τ is time left. This term creates a learnable slope toward the exit before closing, while remaining inactive when the visitor still has enough slack to continue sightseeing.

Crowding also had to be handled conservatively. When congestion was penalized too harshly, the art lover learned a degenerate policy that avoided crowded masterpieces altogether. The final full-museum reward therefore has no separate explicit congestion penalty; density discounts art value smoothly rather than dominating it, because a crowded Botticelli room should remain preferable to an empty corridor. Similarly, strong boredom penalties encouraged loops because moving reset the local penalty. The final reward relies primarily on satiation and a small step cost, with only a modest post-satiation boredom term active in the booking wrappers.

RAMA adds a booking-phase reward:

$$r_{\text{book}} = -d_{\text{decline}} n_{\text{decline}} - k_{\text{mm}} \sum_{g \in \mathcal{B}} |w_g - \hat{t}_g| - d_{\text{noshow}} n_{\text{noshow}}.$$

The mismatch term was the decisive booking-phase addition. It charges, immediately at commitment time, the distance between the reserved window w_g and a frozen expected arrival time $\hat{t}_g$. Without this off-pace proxy, the cost of a bad reservation arrives hundreds of steps later, creating a severe credit-assignment problem. We also encode “books the three masterpieces” as a visitor-type trait (`allow_decline=False`); otherwise the availability of a decline action creates a degenerate solution in which the agent avoids delayed risk by reserving nothing.

5 Algorithms and Masking

The full museum is too large for a table. We use PPO-family actor-critic methods for the main agents and DQN as the value-based control. PPO optimizes the clipped surrogate

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad \rho_t = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}.$$

DQN minimizes a squared Bellman error against a target network and replay buffer:

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right].$$

Action masking is the key implementation detail. In the full navigation environment, most graph-action indices are illegal in most states. In the RAMA wrapper, the action space is smaller but still phase-dependent: lead choices are legal on the booking screen, commit choices are legal at the reservation step and stay/move choices are legal during the walk. MaskablePPO sets illegal logits to $-\infty$ before the softmax:

$$\pi_{\theta}(a \mid s) = \frac{\exp(z_a) \mathbb{I}[a \in \mathcal{A}(s)]}{\sum_{b \in \mathcal{A}(s)} \exp(z_b)}.$$

Invalid actions therefore receive exactly zero probability and zero gradient. Unmasked PPO must learn legality indirectly by sampling illegal moves and paying penalties, which wastes data. In easy cells the difference is small; in the packed tourist RAMA cell it is outcome-determining.

The PPO-family arms use Stable-Baselines3 MLP policies with two 128-unit hidden layers, learning rate 3×10^{-4} , rollout length 2048, batch size 256, 10 epochs per update, GAE $\lambda = 0.98$, clip range 0.2 and entropy coefficient 0.01. The no-booking walk baselines train for 150k steps with $\gamma = 0.995$ for art lovers and 0.997 for tourists. RAMA booking policies train for 100k steps with $\gamma = 0.999$, which keeps delayed reservation payoffs alive. DQN uses a 100k replay buffer, target updates every 1000 steps, train frequency 4, learning starts at 2k steps and ε -greedy exploration decaying from 0.30 to 0.05.

Training uses common random numbers for evaluation: competing policies are scored on the same fixed crowd seeds, so differences are due to policies rather than lucky simulation noise. The figures in this paper use five training seeds $\{1, 3, 7, 42, 202606\}$, each evaluated deterministically on six fixed crowd seeds 900000–900005. The repository also stores the selected single-model deliverable evaluation in `results_rl_booking.json`; the figures use the five-reseed means because they reveal instability rather than hiding it.

6 Results

6.1 Booking policy

Table 3 shows the individual RL result used in the figures. The baseline is the best matched no-intervention walk. The intervened score is the MaskablePPO RAMA booking policy in the all-11 world, averaged over five independent training seeds and six fixed evaluation seeds. The art lover at the packed crowd is the one visibly unstable cell: four runs learn the intended 3/3 booking pattern, while one run collapses, producing both the large standard deviation and the 2.6/3 mean check-in rate.

The selected single-model deliverable run is stronger in the art-lover high-crowd cells, reaching 6,297 points at 2,500 visitors and 5,646 at 5,000 visitors in the six-seed deterministic evaluation. The five-reseed table is more conservative and more informative technically, because it exposes the rare but important failure mode. The tourist result is more stable: the shorter checklist and shorter dwell targets make it easier to book the three windows and execute the route. The learned lead-time curve still has the intended comparative static. Art lovers move from 7-day booking at the quiet crowd to 35-day booking at busy and packed crowds; tourists stay at 7 days until the packed crowd, where the reseeded curve moves later as well.

6.2 Algorithm comparison

Figure 3 isolates the algorithmic conclusion. MaskablePPO is best or tied in every cell. Unmasked PPO collapses for the packed tourist in the intervened world, falling to 2,093 points where Mask-

Table 3: RAMA booking agent versus matched no-intervention walk baseline. Points are five-training-seed means; parentheses report the standard deviation of the intervened arm.

Profile	Crowd	Baseline	RAMA	Gain	Check-ins
Art lover	500	5,112	7,113 (0)	+39%	3.0/3
Art lover	2,500	4,780	6,167 (291)	+29%	3.0/3
Art lover	5,000	4,010	4,800 (1,891)	+20%	2.6/3
Tourist	500	2,694	3,686 (0)	+37%	3.0/3
Tourist	2,500	2,708	3,695 (0)	+36%	3.0/3
Tourist	5,000	2,680	3,685 (0)	+38%	3.0/3

RAMA booking: intervened vs matched baseline (5 seeds, mean $\pm$ std; 3/3 masterpieces secured)

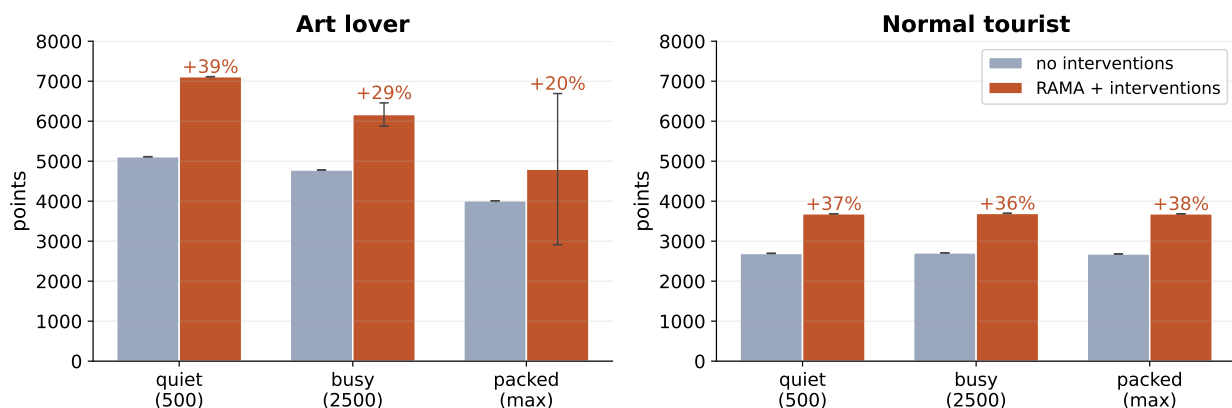


Figure 2: RAMA booking gains over matched no-intervention walk baselines, reported as five-training-seed means with standard-deviation bands. The grid is useful because it shows both profile heterogeneity and crowd-level dependence.

ablePPO reaches 3,685. DQN is competitive in several easy tourist cells but crowd-fragile for the art lover; under the no-intervention baseline, its art-lover return falls from 4,527 at the quiet crowd to 2,044 at the packed crowd in the five-reseed mean.

This pattern is consistent with the algorithms’ failure modes. DQN inherits a maximization step over noisy value estimates, the same issue that motivates Double-Q in the tabular setting. PPO avoids that particular max-over-actions target, but unmasked PPO still spends exploration budget discovering actions that are impossible in the current graph or booking phase. MaskablePPO removes that burden by matching the policy support to the legal action set at every minute.

7 Limitations

The model is only as good as its calibration. Room topology and opening hours are grounded in public sources and the official floor plan, but several behavioral parameters are explicit assumptions: the exact 60/30/10 profile mix, heterogeneity scale, willingness-to-pay elasticity and intervention compliance rates. The results should therefore be read as a controlled simulation study, not as a claim that the real Uffizi would mechanically realize the same percentages.

There is also a reward-design limitation. We use a consistent welfare proxy, but museum experience is not a single scalar. The reward values art exposure, crowd comfort, completion and timely exit; it does not directly model learning, fatigue psychology, social groups, fairness constraints, or curatorial goals beyond room importance and enrichment. Finally, the learned policies depend

Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean $\pm$ std)

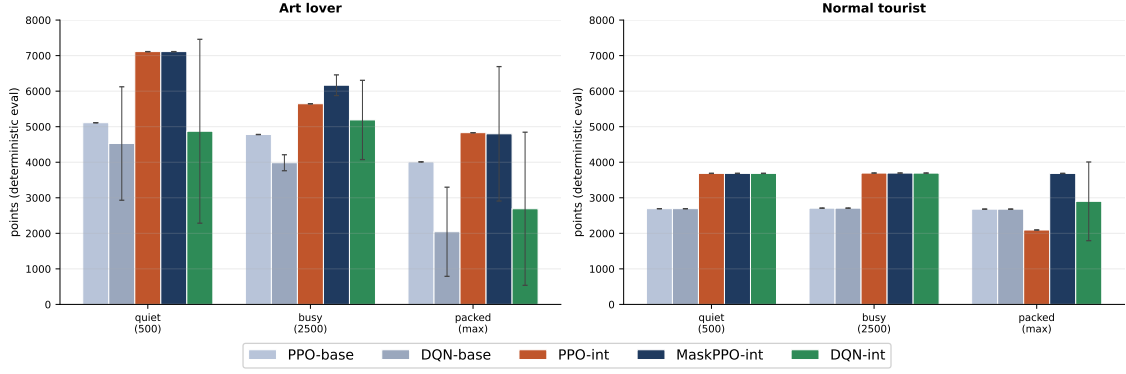


Figure 3: Algorithm comparison across profile and crowd cells. The dark-blue bars are MaskablePPO. The gap to unmasked PPO at the packed tourist cell is the cleanest empirical evidence for action masking.

on the observation shape. Changing the 8-dimensional walk state or 25-dimensional RAMA state invalidates the saved models and requires retraining.

8 Conclusion

The project’s main technical lesson is that useful learning required the right abstraction of the decision problem. The status-quo visitor mostly faces a walking and dwell-time problem. The intervened visitor faces a reservation problem with delayed payoffs, irreversible commitments and legality constraints. Once the MDP exposes those variables, MaskablePPO learns the intended policy: reserve masterpieces early, reserve even earlier on crowded days and pace the visit to collect the booked windows.

At the museum level, the all-11 intervention world reduces the worst density bands and raises completed welfare by 31% at the 5,000-visitor crowd while preserving revenue. At the individual level, the RAMA agent improves over optimized no-intervention baselines for both profiles and all crowds, with one high-variance art-lover cell that is visible only when the five training reseeds are kept. The most robust engineering takeaway is action masking: respecting the legal action set matters more than extra tuning once the environment becomes hard.

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